

MGT403: Advanced Financial Risk Management

Master's in Finance 2025-26, UCSD

MON 8:00AM-10:50 AM PST

LOCATION: WFH 4N128

PROFESSOR: Rossen Valkanov

EMAIL: rvalkanov@ucsd.edu

PHONE: (858) 534-0898

OFFICE: 4S139, 4th Floor Otterson Hall

OFFICE HOURS: Thursdays, 12 pm-1 pm in 4S139 (or ask for a Zoom link)

TEACHING ASSISTANT: Jianbo Bin (Jianbo.Bin@rady.ucsd.edu), Kanji Suzuki (k4suzuki@ucsd.edu)

COURSE DESCRIPTION

Risk management is about implementing modern financial techniques to evaluate and manage various sources of risk using financial strategies and instruments. This class will use class lectures, case studies, book readings, and additional "last minute" handouts from the financial press to illustrate the basic principles of modern risk management.

The Advanced Risk Management course is designed primarily for Masters of Finance students. We will first discuss if, when, and how risk management contributes to firm value. We will analyze various measures of risk--such as Value-at-Risk (VAR) and Cash-Flow at Risk (CAR)--and how financial and non-financial companies use these measures in project selection, performance evaluation, and managerial compensation. Then, we will turn to methods of changing a firm's exposure to financial risk using derivatives. More specifically, we will cover dynamic hedging, the management of exchange rate risk, commodity price risk, interest rate risk, credit risk, weather risk, and operational risk. We will also analyze how strategic risk management can be applied to extract embedded options. A section of the class will cover Basel II and other current banking regulations. Finally, we will discuss cases where financial risk management is vital for the success of a business strategy.

All covered topics will have a significant quantitative component; students will have to work with financial data and will be required to implement the methods discussed in class. The focus of the course is on implementing risk management techniques rather than on derivative pricing. We will rely heavily on case studies to solidify and apply our knowledge to problems encountered in the business world.

Risk management is a very rapidly growing field. If you are interested in risk management, you might want to join the Global Association of Risk Professionals (GARP). GARP offers an official financial risk management (FRM) accreditation (similarly to CPA for accountants and CFA for financial analysts.) The class will use some of the GARP recommended books (see below).

I will assume that you are familiar with basic statistical concepts (means, variances, covariances and linear regression, etc.) and Excel.

Prerequisites:

- Investments

- Empirical methods and financial econometrics

SCHEDULE

Class meets once a week on **Mondays from 8:00AM-10:50AM** in **WFH 4N128**. Two classes fall on university holidays and will be rescheduled as either in-person or remote (see schedule). If the classes are rescheduled as remote, the Zoom link will be on the Canvas website.

The recorded version of the class will be available on Canvas a few hours after the lecture.

Office hours will be held in person or via Zoom. If you cannot come in during office hours, please e-mail me for an appointment. I welcome questions via email.

CANVAS

You must already be familiar with Canvas, but if not, please familiarize yourself with it. Each week, I will post all materials for the class –lecture slides, assignments, quizzes, additional material, Python code and others — in a Module.

COURSE MATERIAL

The weekly lecture notes will be available on Canvas, posted the morning of the lecture.

There is one required book as well as a course reader.

- Book: Philippe Jorion, Value at Risk, 3rd Edition, McGraw-Hill, 2006.
- Course Reader: Available on Canvas (Study.net)

We will cover a few case studies from the course reader and/or additional reading material each week. Please keep up with the readings. If you have trouble accessing the material from the course reader, please let me know as soon as possible.

ASSIGNMENTS AND GRADING

There will be bi-weekly case studies. You have to work in assigned groups. The case studies will be based on class discussions. Please write the answers to the case studies in a memo format of no more than two pages. The limit of two pages is for the typed part only. Please, attach all supporting materials (graphs, tables, pictures, etc.) at the end of the document. You should refer to the supporting materials in the main body of the memo.

We will discuss answers to the case studies at the beginning of the following week's class. You should be ready to take active participation in these discussions.

There will be no midterm exams. There will be a final exam. It will cover class lectures, case studies, readings assigned in the textbook, additional readings, and materials distributed during class. The final exam will be comprehensive and should be sole authored. No collaboration of any kind is allowed.

The grading policy is as follows:

1. Case Studies/Written Exercises (35%) — work in pre-assigned groups. One written assignment per group to be turned in on time.
2. Weekly Quizzes (15%) — We will assign weekly quizzes after each lecture on Canvas. They will help you keep up with the material.
3. Class Participation (10%) — Weekly attendance and class participation, contributing to a productive learning environment.
4. Final Examination (40%) — The final examination will be administered during Finals week.

To prevent one bad performance from affecting your grade, I will drop the lowest case study exercise and weekly quiz grade.

DETAILED SYLLABUS

We **may deviate** from the schedule depending on the pace the class feels comfortable with.

Class 1: 1/05 Intro to Risk-Management; Market Risk; Hedging; Diversification; Value-at-Risk; Cashflow-at-Risk
Reading: Jorion, Ch 4 and 5.

- *Why Manage Risk? HBS 9-294-107
- *Rethinking Risk Management, Stulz, Journal of Applied Corporate Finance
- *A Comparables Approach to Measuring Cashflow-at-Risk for Non-Financial Firms
- *A Framework for Risk Management, Froot, Scharfstein, and Stein, Harvard Business Review

Class 2: 1/12 Finish VAR; Var of a portfolio. Review of Derivatives: Forwards, Futures, Options. Their use in risk-management
Reading: Jorion, Ch 6 and 7.
An Overview of Value-at-Risk (Duffie and Pan), The Journal of Derivatives
*How Accurate are Value-at-Risk Models at Commercial Banks (Berkowitz and O'Brien)
*Fact and Fantasy in the Use of Options, Fischer Black, Financial Analyst Journal
*Leland, O'Brien, Rubinstein Associates Incorporated: Portfolio Insurance, HBS 9-294-061

Class 3: 1/19 **MLK HOLIDAY --- To be rescheduled as in-person or remote**
Finish Futures, Forwards, and Options. Hedging with Options, Dynamic Hedging; Measuring and Managing Exchange Rate Risk
Reading: *Replicating Options with Positions in Stock and Cash, Rubinstein and Leland, Financial Analyst Journal
*Identifying, Measuring, and Hedging Currency Risk at Merck, Lewent, Kearney, Journal of Applied Corporate Finance
*The New Risk Management: The Good, the Bad, and the Ugly (Dybvig and Marshall), Federal Reserve Bank of St. Louis.
*Credit General, SA, HBS 9-296-011

Class 4: 1/26 Finish Market and Exchange Rate Risk; Measuring and Managing Weather Risk
Reading: *Aspen Technologies, Inc: Currency Hedging Review, HBS 9-296-027
*AIG-Stanford GSB, Case A-203.
*Managing Foreign Exchange Risk with Derivatives (Brown)

Class 5: 2/02 Hedging and Financial Institutions
Reading: *The Collapse of Metallgesellschaft: Unhedgeable risks, Poor Hedging Strategy, or Just Bad Luck? Edwards and Canter, Journal of Futures Markets.
The Exchange Rate Exposure of U.S. and Japanese Banking Institutions (Chamberlain, Howe, Popper), Wharton School Working Paper.

Class 6: 2/09 Measuring and Managing Commodity Price Risk
Reading: *American Barrick Resources Corporation: Managing Gold Price Risk; HBS 9-293-128

*Note on Crude Oil and Crude Oil Derivative Markets

Class 7: 2/16

Presidents' Day HOLIDAY --- To be rescheduled as in-person or remote

Measuring and Managing Liquidity Risk; Measuring and Managing Interest Rate Risk

Reading:

*Jorion, Ch 14

*Long-Term Capital Management, L.P. (A,B,C,D), HBS 9-2000-007, 8, 9, 10

Class 8: 2/23

Risk Measuring and Managing Credit Risk

Reading:

Jorion, Ch 13

*An Overview of Credit Derivatives, HBS 9-297-086

*Enron Corp-Credit Sensitive Notes, 9-297-099

*Orange County, Jorion's website.

Class 9: 3/02

Strategic Risk Management: Use of Financial Engineering

Reading:

*MW Petroleum Corporation (B), HBS 9-295-045

Class 10: 3/09

Operational Risk; Political and Regulatory Risk; Integrated Risk Management

Reading:

Jorion, Ch 19, 20, 21, 22

*Amendment to the Capital Accord to Incorporate Market Risks, Basle Committee on Banking Supervision 1996.

*Operational Risk Management, Basle Committee on Banking Supervision 1998.

Final Exam: 3/16

In-person. Time: 8am- 11am PST. Location: 1E106/107

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind. The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://www.senate.ucsd.edu/manual/appendices/app2.htm#AP14>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member. The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.